

Measuring School Facilities from Aerial Imagery

1. Validation

Using a fixed random seed, I selected eight of the 25 schools (almost one-third of the dataset) for validation. Reference values were established independently using Google Earth/Street View (for fencing) and official school websites. For all 25 schools, I generated two sets of measurements from NAIP imagery: manual labels with confidence scores and Qwen 3.6 27B predictions. For each validated attribute and each methodology, the exact-match error rate was calculated as the number of incorrect measurements divided by eight. Across 48 school-attribute comparisons, the overall error rate was 4.2% for hand labeling and 18.8% for Qwen. Complete field-level error rates and validation data are available in the GitHub repository.

2. Failures

Hand-labeled values were obtained for all 25 schools while Qwen omitted 4% of the total entries. However, as anticipated in my answer for the first part of the assignment, fencing presence, type, and perimeter could not be measured reliably because fence lines were too narrow for the imagery resolution and were often obscured by trees or shadows. These fields were marked as unassessed rather than guessed.

Solar-panel presence was sometimes visible, but panel area could not be estimated correctly because of unclear roof boundaries.

Qwen struggled to differentiate between hard courts and full-sized fields in some cases due to low resolution. This must be taken care of during fine-tuning. Groq rate limits also required inference to be completed in several passes, although all schools were eventually processed.

3. Cost

The five-hour limit was mainly spent developing and testing the approach, so I exclude it from the per-school estimate. Once the pipeline is set up, I estimate approximately two minutes for automated imagery retrieval and Qwen inference, followed by three minutes of human review (about five minutes per school). At this rate, 130,000 schools would require roughly 10,800 processing hours, although automated steps could run in parallel.

Groq inference would cost roughly \$0.007 per school, or \$950 total; including cloud compute, retries, and storage, I estimate an initial technical cost of \$1,200–\$1,500, plus approximately \$130 per month for storing the imagery. However, this can be reduced by taking appropriate steps.

4. What I would do with 100 hours

- Evaluate higher-resolution imagery sources and use closer campus crops to improve detection of small facilities.
- Supplement the aerial imagery with publicly available Street View images and school campus-tour videos to assess fencing and indoor facilities such as pools and courts. Given the five-hour constraint, the current analysis was limited primarily to exterior facilities visible in aerial imagery.
- I would build a larger labeled dataset and test models such as Grounding DINO and SAM to see whether they detect the facilities more accurately. Based on the results, I would fine-tune the best model and use it to build a more automated pipeline, while still manually reviewing uncertain cases.

5. Tooling

- Google Colab/Python:** Developed and executed the pipeline.
- OpenStreetMap and Esri:** Verified school coordinates and campus locations.
- NAIP via Microsoft Planetary Computer:** Acquired aerial imagery.
- Qwen 3.6 27B via Groq:** Generated automated facility predictions.
- Google Earth/Street View and official school websites:** Established reference values and reviewed historical changes.
- ChatGPT/Codex:** Assisted with code development, debugging, and documentation; all outputs were manually reviewed.